Reg. No. | | | | | | | | | | |

[17EE45]

FOURTH SEMESTER B.Tech EXAMINATION, JUNE–2019

TRANSFORMERS AND INDUCTION MACHINES

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions selecting at least **TWO** questions from each part.
2. Any missing data may be suitably assumed.

PART - A

1. a. Explain in brief the concept of self-inductance and mutual inductance. **04**

   b. Derive the EMF equation of single phase transformer considering the voltage also. **04**

   A 125KVA transformer having primary voltage of 2000V at 50Hz has 182 primary and 40 secondary turns. Neglecting losses, calculate **06**

   i. the full-load primary and secondary currents,

   ii. the no-load secondary induced e.m.f. and

   iii. the maximum flux in the core.

2. a. Draw and explain the phasor diagram of the single phase transformer for leading lagging power factor when loads are connected to the secondary and also observe the flux with respect to primary and secondary. **06**

   b. Find the active and reactive components of no-load current, and the no-load current of a 440/220V single-phase transformer if the power input to the hv winding is 80W. The low-voltage winding is kept open. The power factor of the no-load current is 0.3 lagging. **04**

   c. A 230/110 V single-phase transformer takes an input of 350VA at no load and rated voltage. The core loss is 110W. Find; **04**

   i. the iron-loss component of no-load current,

   ii. the magnetizing component of no-load current and

   iii. no-load power factor.

3. a. Describe in details open circuit and short circuit test on transformer determine the circuit constants, efficiency and regulation without actually loading the transformer. **07**

   b. A 100KVA, 50HZ, 440/11000V, 1-phase transformer has an efficiency of 98.5% when supplying full load current at 0.8 power factor lagging and an efficiency of 99% when supplying half full-load current at unity power factor. Find the core losses and the copper losses corresponding to full-load current. At what value of load current will the maximum efficiency be attained? **07**

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4. a. How many types of possible connections for three phase transformer bank can be done using single phase transformer? What are the factors affecting the choice of connections? Explain in details star delta connection. **08**

b. An 11000/440V, 50Hz, 3-phase transformer is delta connected on the HV side and the LV winding are star connected. There are to be 12V per turn and the flux density is not to exceed 1.2 Wb/m^2. Calculate the number of turn per phase on each winding and the net iron cross-sectional area of the core. **06**

PART - B

5. a. A 50 kw, 6 POLE, 50Hz, 450V 3-phase slip ring induction motor furnished the following test figures : **14**

No-load test : 450 V, 20A, p.f.=0.15

Block rotor test : 200V, 150A, p.f=0.3

The ratio of stator to stator to rotor copper losses on short circuit was 5 : 4 Draw the circle diagram and determine from it

i. The full load current and power factor,

ii. The maximum torque and the maximum power input,

iii. Slip at full load

iv. Efficiency at full load.

6. a. Explain different mechanisms for the speed control of induction motor. Describe any two methods of speed controls of a three phase induction motor. **10**

What is magnetic locking? Explain in details cogging and crawling. **04**

7. a. Describe the generation of double revolving field theory of single phase induction motor. **06**

b. A 230V, 50Hz, 4 pole single phase induction motor has the following equivalent circuit impedances : **08**

$R_{1m}=2.2\Omega$, $R_2=4.5\Omega$, $X_{1m}=3.1\Omega$, $X_2=2.6\Omega$, $X_m=80\Omega$.

Friction, windage and core loss=40W

For a slip of 0.03 pu, calculate ;

i. input current

ii. power factor

iii. developed power

iv. output power

v. efficiency.

8. a. What are the advantages and disadvantage of autotransformer state the application of auto transformers **06**

b. State the difference between power and distribution transformers **04**

c. Explain the principle of operation of constant current transformer and its applications. **04**